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1. What are the differences between Native, Hybrid, and Cross-platform mobile application development, and when is each approach commonly used?

Native, Hybrid, and Cross-platform are three different approaches to mobile application development. Native development creates apps specifically for one operating system, such as Android or iOS, using platform-specific programming languages. It offers the best performance and full access to device features, making it ideal for high-performance applications like games, banking apps, and camera-intensive apps. Hybrid development uses web technologies such as HTML, CSS, and JavaScript wrapped inside a native container, allowing one codebase to run on multiple platforms.

2. What is the Flutter Ecosystem, and how does it help developers create modern mobile applications?

The Flutter Ecosystem is the collection of tools, technologies, and resources that support Flutter app development. It includes the Flutter framework, the Dart programming language, development tools, widgets, plugins, packages, and a large developer community. Together, these components help developers create modern, responsive, and visually appealing mobile applications more efficiently.

3. What is the Software Development Life Cycle (SDLC), and why is it important in building successful applications?

The Software Development Life Cycle (SDLC) is a structured process used to develop software from start to finish. It consists of several phases:

Planning

Requirements Analysis

System Design

Development (Coding)

Testing

Deployment

Maintenance

SDLC is important because it:

Provides a clear roadmap for development.

Improves software quality through proper testing.

Reduces risks, errors, and development costs.

Helps teams work efficiently together.

Ensures the final application meets user and business requirements.

4. How do Flutter development practices and the SDLC work together in creating real-world mobile applications?

Flutter development practices align closely with the stages of the SDLC. During the planning and requirements phases, developers determine the application's goals and decide whether Flutter is the best framework for the project. In the design and development stages, Flutter's reusable widgets and Hot Reload feature allow developers to create and improve user interfaces quickly.

5. Why should developers understand these concepts before starting an application project?

As Developers we should understand these concepts before starting a project because they provide the foundation for making informed decisions throughout the development process. Knowing the differences between native, hybrid, and cross-platform development helps in selecting the most suitable approach based on the project's goals, budget, and target users.